LESSON 21

ESP32 SUPER STARTER KIT · 2-HOUR SESSION

My Invention

Plan, build, test and show off a project of your own design.

Name: _____ Date: _____

TODAY YOU WILL LEARN

- How real engineers go from an **idea** to a working **prototype**
- How to combine **inputs** and **outputs** from earlier lessons
- How to build and test **one piece at a time**
- How to explain your invention to others

SESSION PLAN

0:00	The inventor's process + idea menu
0:15	Example: the Smart Room Monitor
0:30	Fill in your planning sheet
0:50	Short break
0:55	Build and test, one piece at a time
1:40	Show and tell
1:55	Reflection and tidy up

PARTS TO COLLECT

- ☐ ESP32 board + USB-C cable, 2 breadboards
- ☐ Whatever your project needs (see your plan!)
- ☐ For the example: DHT11, PIR sensor, OLED, passive buzzer
- ☐ Your old sketches from lessons 1–20
- ☐ A pencil

1. You are the inventor now

You've learned about LEDs, buttons, sounds, screens, sensors, motors, cards, remotes and Wi-Fi. Today **you** decide what to build!

> **Big idea: the inventor's loop**
>
> Real engineers don't build everything at once. They follow a loop: **Idea** → **Plan** → **Build a small piece** → **Test** → **Fix** → **Build the next piece…** and finally **Show**. If something goes wrong, that's normal. It's just the *Test* → *Fix* part of the loop.

Every invention is made of **inputs** (things that sense the world) and **outputs** (things that do something):

Inputs (sense)	Outputs (do)
Button, keypad, knob, joystick, tilt switch	LED, RGB LED, 7-segment display
Light sensor, temperature (DHT11), thermistor	Buzzer: beeps and tunes
Distance (ultrasonic), motion (PIR), tilt (GY-6500)	OLED screen: words, numbers, pictures
RFID card, IR remote, Wi-Fi web page	Servo, stepper motor, fan motor

2. Idea menu

Pick one, change one, or invent your own!

Idea	What it does	Main parts	Lessons to look back at
Smart Room Monitor	Shows temperature, warns when too hot, chimes when someone comes in	DHT11, PIR, OLED, buzzer	10, 11, 13
Secret Treasure Box	Opens a servo lock with the right card *or* keypad code	RFID or keypad, servo, LEDs	14, 16, 17
Parking Helper	Beeps faster and shows distance as a car gets closer	Ultrasonic, buzzer, OLED or LEDs	6, 10, 12
Reaction Game	Two players race to press their button after a light	Buttons, LEDs, 4-digit display	3, 9
Plant Guardian	Warns when it's too dark or too hot for a plant	Photoresistor, DHT11, RGB LED	4, 5, 11
Desk Fan Controller	Fan speeds up when it's warmer; remote to turn it off	DHT11, DC motor, IR remote	11, 15, 18
Phone-Controlled Disco	Pick colours and tunes from a web page	RGB LED, buzzer, Wi-Fi	5, 6, 20
Intruder Alarm	Arm with a keypad code; motion sets off a siren	PIR, keypad, buzzer, LEDs	13, 16

3. Worked example: Smart Room Monitor

Here is one finished project, so you can see how the pieces come together. You can build it as it is, or use it as a starting point and change it.

✓	Part	Pin	Power
☐	DHT11 **S** (signal)	**4**	+ to **3V3**, – to **GND**
☐	PIR **OUT** (middle)	**27**	VCC to **5V**, GND to **GND**
☐	Buzzer **+**	**33**	– to **GND**
☐	OLED **SDA / SCL**	**21 / 22**	VCC to **3V3**, GND to **GND**

Type this in → save it as **L21_SmartRoom** 62 lines

```
// Lesson 21: Example project – Smart Room Monitor
// DHT11 on 4, PIR on 27, buzzer on 33, OLED on 21/22.

#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>

const int DHT_PIN = 4;
const int PIR_PIN = 27;
```

```
const int BUZZER_PIN = 33;
const float TOO_HOT = 28.0;

Adafruit_SSD1306 display(128, 64, &Wire, -1);
DHT dht(DHT_PIN, DHT11);

float temp = 0;
float humidity = 0;
unsigned long lastRead = 0;
bool wasMotion = false;

void setup() {
  pinMode(PIR_PIN, INPUT);
  dht.begin();
  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  display.setTextColor(WHITE);
}

void loop() {
  // 1. Read the DHT11 every 2 seconds (it is slow)
  if (millis() - lastRead > 2000) {
    temp = dht.readTemperature();
    humidity = dht.readHumidity();
    lastRead = millis();
  }

  // 2. Ding-dong when someone walks in
  bool motion = digitalRead(PIR_PIN) == HIGH;
  if (motion && !wasMotion) {
    tone(BUZZER_PIN, 1200, 150);
    delay(200);
    tone(BUZZER_PIN, 900, 300);
  }
  wasMotion = motion;

  // 3. Show everything on the screen
  display.clearDisplay();
  display.setTextSize(2);
  display.setCursor(0, 0);
  display.print(temp, 1);
  display.print(" C");
  display.setCursor(0, 20);
  display.print(humidity, 0);
  display.print(" %");
  display.setTextSize(1);
```

```
  display.setCursor(0, 44);
  if (temp > TOO_HOT) display.print("Too hot! Open a window");
  display.setCursor(0, 56);
  if (motion) display.print("Someone is here!");
  else display.print("Room is empty");
  display.display();
  delay(100);
}
```

Where did each piece come from?

Part of the sketch	Learned in
`DHT dht(DHT_PIN, DHT11);` and `readTemperature()`	Lesson 11: Weather Station
`if (millis() - lastRead > 2000)` (do something every 2 s without `delay`)	Lesson 9
`digitalRead(PIR_PIN)` and remembering `wasMotion`	Lesson 13: Motion Alarm
`if (motion && !wasMotion)`: only beep at the **moment** motion starts	Lesson 3 (buttons)
`display.print(temp, 1)`: show 1 decimal place	Lesson 10: OLED

Notice the comments

The `loop()` is split into steps: **1. read**, **2. decide**, **3. show**. Most inventions follow this pattern. It's a great way to plan your own sketch!

4. My planning sheet

My invention is called:

What it does (one or two sentences):

Who would use it, and why?

Inputs and outputs:

Inputs (what it senses)	Outputs (what it does)

Parts list and pins:

✓	Part	ESP32 pin(s)	Power (3V3 or 5V)	Lesson to look back at
☐				
☐				
☐				
☐				
☐				

Pin check before you wire

Use pins from your earlier lessons. **Don't use** pins 0, 2, 12 or 15. Pins 34–39 are **input only**. Two parts can't share one pin, except I²C parts (OLED and GY-6500) on 21/22. 5 V parts like the servo and ultrasonic sensor get power from **5V**, but the ESP32 pins only take 3.3 V!

Sketch plan: what will `loop()` do? Write it in plain words first.

Step	What my sketch does
1. **Read** the inputs	
2. **Decide** what to do	
3. **Show / do** something	

5. Build and test, one piece at a time

The golden rule

Never wire everything and write everything at once. Add **one** part, get it working, then add the next. When something breaks, you'll know it was the last thing you added.

- ☐ Step 1: wire your **first input** and print its value to the Serial Monitor
- ☐ Step 2: wire your **first output** and test it on its own
- ☐ Step 3: connect them with an `if`: *when input does X, output does Y*
- ☐ Step 4: add the next part… and test again
- ☐ Step 5: make it fun: sounds, messages, animations

Test log: write down what you tried, especially when it didn't work!

What I tried	What happened	What I'll change

Stuck?

1. Open the sketch from the lesson that used the part, and check the wiring and code against it.
2. Add `Serial.println(…)` to see what your variables really are.
3. Explain your code out loud to a friend (or a rubber duck!). You'll often spot the problem yourself.

6. Show and tell

Each inventor gets **2 minutes**. Use these prompts:

- ☐ What is it called and what problem does it solve?
- ☐ Demo: show it working!
- ☐ What was the hardest bug, and how did you fix it?
- ☐ What would you add if you had another 2 hours?

Extension: level it up

Finished early? Add a **Wi-Fi web page** (Lesson 20) so your invention can be checked or controlled from a phone. Your teacher has an example that adds this to the Smart Room Monitor.

7. Reflection

Look back

1. Which lesson was your favourite, and why?

   _____

2. What's one thing you can do now that you couldn't do in Lesson 1?

   _____

3. Name one invention you'd love to build next, maybe with parts that aren't in the kit.

   _____

   _____

Tidy up: take a photo of your project! Then carefully put every part back in its place in the kit.
Congratulations, inventor! 🎉